

# David Bean

## DATA SCIENCE, DATA PIPELINES

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### SKILLS

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|-------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>        | Python, SQL, Bash, R                                                                                        |
| <b>Acquisition</b>      | REST APIs, web scraping, JSON, requests, BeautifulSoup, web crawlers, CSV, geospatial data, NetCDF          |
| <b>Wrangling</b>        | validation suites, merging sources, deduplication, group-by aggregation, sanity tests                       |
| <b>Libraries</b>        | pandas, NumPy, SciPy, GeoPandas, dplyr, xarray, subprocess                                                  |
| <b>Pipelines</b>        | incremental processing, resumable task queues, checkpointing, metrics logging, email alerts on failure      |
| <b>Scale</b>            | parallelization, larger-than-memory processing (100+ GB), runtime profiling                                 |
| <b>Linux</b>            | parallel batch jobs, SSH, Slurm, grep, sed, awk, Make, named pipes, tmux                                    |
| <b>Machine Learning</b> | scikit-learn, XGBoost, feature selection, cross-validation, gradient boosting, PCA, k-means, UMAP           |
| <b>Statistics</b>       | hypothesis testing, t-tests, chi-square, non-parametric tests, multiple-comparison correction, effect sizes |
| <b>Modeling</b>         | linear regression, logistic regression, experimental design, confidence intervals, correlation analysis     |
| <b>Visualization</b>    | interactive dashboards, matplotlib, Google Sheets, Excel, Jupyter, ggplot2, Vega-Altair, RStudio            |
| <b>Databases</b>        | schema design, joins, DDL, ER diagrams, GROUP BY, primary and foreign keys                                  |
| <b>Version Control</b>  | Git, GitHub, pull requests, GitLab, branching and merging                                                   |
| <b>Containers</b>       | Docker, Dockerfiles, GPU containers, Podman, Apptainer, venv                                                |

### EDUCATION

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#### Master of Science in Biomedical Informatics

May 2026

University of Utah | Salt Lake City, UT | Emphasis: Data Science and AI | GPA: 3.95

#### Bachelor of Science in Biology, magna cum laude

May 2023

Utah Tech University | Saint George, UT | Emphasis: Biomedical Sciences | GPA: 3.80

### EXPERIENCE

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#### Research Assistant II

Jun 2025 - Dec 2025

University of Utah - Department of Biomedical Informatics, Salt Lake City, UT

- Diagnosed why a published mutation classifier's precision fell from 0.71 on its benchmark data to 0.02 on pediatric patient data, showing its performance was highly sensitive to class ratio (somatic mutations were 18.7x rarer)
- Optimized pipeline efficiency and stability, reducing whole-genome analysis runtime 10x (~50 to 5 hours) via parallel, containerized per-chromosome Slurm jobs and a custom dynamic task scheduler that resumes from last completed step
- Eliminated harmless mutations from patient data by developing a pipeline that used set operations to filter against population databases (1000 Genomes, gnomAD), while retaining 96% of known cancer mutations
- Guided feature selection with dimensionality reduction (PCA, UMAP), then benchmarked GBMs (CatBoost, LightGBM, XGBoost) through a standard train, tune, resample workflow and selected the model by leave-one-out cross validation
- Influenced the project's direction in weekly technical presentations that summarized methods and findings with data visualization and statistical analysis, and collaborated with domain experts (cancer genomics, data science) on next steps
- Defended investment in the research to department leadership via a 30-minute technical talk, distilling 6 months of work and arguing for continued support of the lab's aims

#### Research Assistant

Oct 2023 - Apr 2024

Utah Tech University - Klabacka Lab, Saint George, UT

- Automated retrieval of protein data for 4,747 genes across 5 species by REST API in a Dockerized Bash and Python pipeline that cleans, processes, and ranks each gene by genetic similarity between species, extensible to related bacteria
- Batch-generated data visualizations in R (ggplot2), looping one reusable chart design over data from 141 bacterial species to map a protein at 5 taxonomic levels and highlight positions where over 98% of species share the same amino acid
- Demonstrated that the target protein, though vital to bacterial survival, was only average in genetic similarity between species (56% of 2,257 genes scored higher), suggesting its few unchanging positions are what cells cannot live without

- Collaborated with a 5-person research team through a shared GitHub repository and regular lab meetings, unified code from multiple contributors into one automated pipeline, and presented findings at 3 research poster sessions

### Laboratory Instructor, Biology 1625

Aug 2023 - May 2024

Utah Tech University - Department of Biological Sciences, Saint George, UT

- Led 3 lab sections of 18 to 24 students through full research cycles, teaching them to pose a question, state formal null and alternative hypotheses, design experiments, draw conclusions from statistical tests, and present for peer feedback
- Lectured on fundamentals of statistical testing and data analysis, including choosing the right types of tests for a dataset (paired and unpaired t-test, chi-square, linear regression) and visualizing findings in Google Sheets and Excel
- Trained students to combine statistical tests with multiple visualizations, and went beyond the syllabus with willing teams (KDE plots, post-hoc tests) to reveal patterns that only emerge when data is viewed from more than one perspective

### Academic Tutor

Jan 2023 - May 2024

Utah Tech University - TRiO Student Support Services, Saint George, UT

- Provided first-generation and non-traditional students with a structured approach to independent study, assessing their knowledge and teaching them how to close the gaps in calculus, trigonometry, statistics, biology and chemistry
- Modeled programming concepts through worked examples in one-on-one sessions on introductory Python and R, building an iterative, experimental instinct for solving problems that students could apply to their own code

### PROJECTS

#### Interactive Monarch Migration Map

2024

- Told the story of monarch migration in an interactive Vega-Altair dashboard of sightings and weekly temperature maps
- Processed climate temperature grids (ERA5, NetCDF) with xarray, tuning their resolution in space and time
- Harmonized 227,000+ monarch sightings with the temperature grids by reprojecting both to the Mercator projection

#### Motorcycle Crash Risk Analysis

2025

- Backed all 14 group comparison charts with hard numbers, a statistical test and p value behind each (Cliff's delta 0.75)
- Merged 7 years of federal crash files on a composite key into 870,000+ person-level records with pandas
- Documented the full analysis in a 20-page report, from data completeness checks to the limits of the models
- Reported the limits of models predicting a sparse positive class (3.8% fatal), where 92% accuracy says little